

LoPoS TDoA scheduling — week-over-week localization yield

ilvo-gw gateway · report generated 2026-06-08

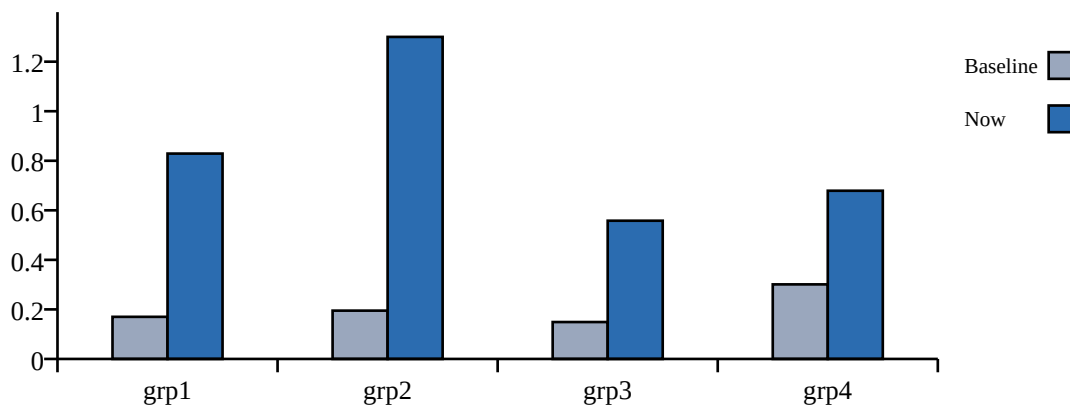
Periods compared

Window	Dates (local)	Duration	System state
Baseline (one week earlier)	Sun 31 May 09:48 → Mon 1 Jun 08:59	23 h 11 m	Pre-proactive blind round-robin cell selection
Since last deploy	Sun 7 Jun 09:48 → Mon 8 Jun 08:59	23 h 11 m	Full current stack (commit 03d74dd)

Both windows: identical clock window (09:48→08:59), full 24-hour data coverage with no gaps, same Sunday→Monday weekday alignment, and the same fleet (~17 active tags per group). Metric = per-tag valid fixes per minute (position rows with ≥ 3 hyperbolas, per group, divided by window minutes and by distinct tags).

Result

Group	Baseline fixes	Baseline ptag/min	Now fixes	Now ptag/min	Change
grp1	4,013	0.170	19,600	0.829	+388%
grp2	4,614	0.195	30,738	1.300	+566%
grp3	3,525	0.149	12,415	0.558	+274%
grp4	7,125	0.301	16,066	0.679	+125%
TOTAL	19,277		78,819		+309%



Per-tag fixes per minute, by group (higher is better).

What changed since 1 June (high level)

- Built a cell-curation toolchain that re-derives TDoA cells from **measured** UWB reception (reception count, not RSSI).
- Introduced **proactive, group-aware cell selection**: each tag is localized in the cell nearest its own recent position instead of blind round-robin — rolled out system-wide.
- Cell choice and the lost-tag fallback now use **centre-of-mass** geometry (tag vs. cell anchor centroids); held multi-round trials for low-signal tags.
- Added a **moving TDoA window** that slides the measurement block through the hyperframe each cycle — eliminates the stale-sync mismatch storm and the “first-report-wins” bundle rejections.
- Fixed cold-start scheduling after a service restart (no more wasted superframes on a non-existent cell).

- Made tag liveness symmetric: a tag with no recent status report is dropped from localization (no slots wasted on absent devices) and re-admitted automatically on its next report.

Interpretation

This is a **week-over-week** comparison: the baseline pre-dates the entire programme, so the +309% reflects the cumulative effect of all the changes above, not a single deploy. Both windows are activity-matched (same weekday, same clock hours, same fleet), and the gain is consistent across all 24 hours, so it is a robust system-level improvement rather than a diurnal artefact.